

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18 June 2026
Student Name	Narendula Jashwanth
Project Name	ShopEZ – Full Stack E-Commerce Platform Using MERN Stack
Maximum Marks	4 Marks

Technical Architecture:

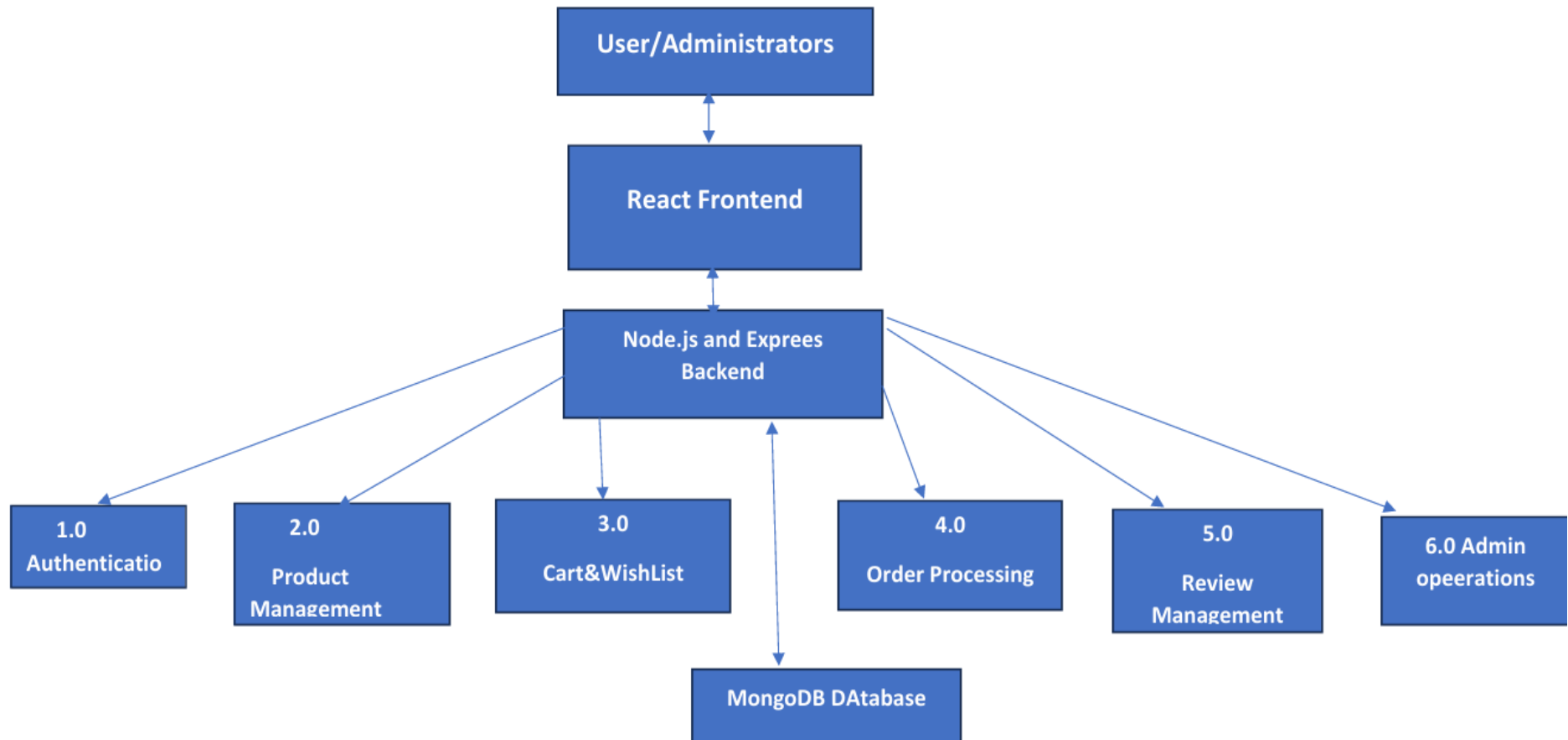


Figure 1: Technology Architecture of the shopEZ

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	The user interacts with the application through web pages such as login, register, products, cart, orders, profile, and admin dashboard.	React.js, Vite, HTML, CSS, JavaScript
2.	Application Logic-1	Handles user registration, login, authentication, and profile management.	Node.js, Express.js, JWT, bcrypt
3.	Application Logic-2	Handles product listing, product details, adding, updating, and deleting products.	Node.js, Express.js
4.	Application Logic-3	Handles cart operations such as add to cart, update quantity, remove item, and calculate total.	Node.js, Express.js
5.	Application Logic-4	Handles order placement, order history, order status, and admin order management.	Node.js, Express.js
6.	Database	Stores user details, product details, cart data, and order details.	MongoDB, Mongoose
7.	File Storage	Stores or manages product images used in the product section.	Local File System / Image URL
8.	External API-1	No external API is used in the current version. Future scope may include payment gateway API.	Not Applicable / Future Payment API
9.	External API-2	No second external API is used currently. Future scope may include delivery tracking API.	Not Applicable / Future Delivery API
10.	Machine Learning Model	No machine learning model is used in the current version of ShopEZ.	Not Applicable
11.	Infrastructure (Server / Cloud)	Application runs on local system during development. It can be deployed on cloud in the future.	Localhost, Node Server, MongoDB, Vercel/Render/Future Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The project uses open-source technologies for frontend, backend, and database development.	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	User authentication is implemented to protect login access. Passwords can be encrypted before storing in the database.	JWT Authentication, bcrypt Password Hashing
3.	Scalable Architecture	The project follows a three-tier architecture, so frontend, backend, and database can be improved or scaled separately.	React Frontend, Express Backend, MongoDB Database
4.	Availability	The application can be accessed locally during development and can be hosted online for wider availability.	Localhost / Cloud Deployment
5.	Performance	The application is designed with simple pages, API-based data transfer, and organized database collections for better response time.	REST APIs, MongoDB Queries, React Components

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>